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科目：

## 作業 1

(1)  $V_{od} = V_{DS, sat} = 0.5V$

(2) 因為  $V_{DS} > V_{DS, sat}$ ，所以 NMOS 管進入恆流導通區  
若考慮厄利效應，則

$$I_D = \beta_n V_{od}^2 \left(1 + \frac{V_{DS}}{V_E}\right) = 2 \times 10^{-3} \times 0.5^2 \times \left(1 + \frac{1}{50}\right) = 5.1 \times 10^{-4} A = 0.51 mA$$

若不考慮厄利效應，則

$$I_D = \beta_n V_{od}^2 = 0.5 mA$$

(3) 若考慮厄利效應，則

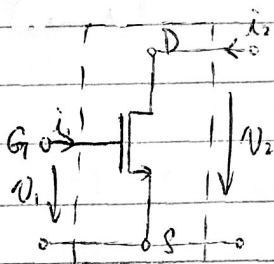
$$I_D = \beta_n V_{od}^2 + \frac{V_{DS}}{\frac{V_E}{\beta_n V_{od}^2}} = 0.5 mA + \frac{V_{DS}}{100 k\Omega}$$

即源電流  $I_{D0} = 0.5 mA$ ，源內阻  $R_{D0} = 100 k\Omega$

## 作業 2

NMOS

電路符號  
及  
二端口網絡

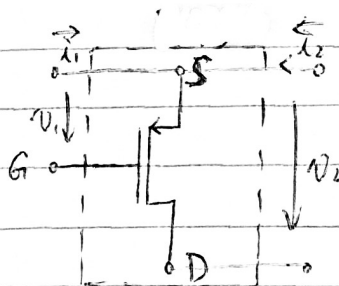


$$\lambda_G = 0$$

元件約束方程

$$I_D = \begin{cases} 0, & V_{GS} < V_{TH} \\ 2\beta_n [(V_{GS} - V_{TH}) V_{DS} - 0.5 V_{DS}^2], & \min(V_{GS}, V_{DS}) > V_{TH} \\ \beta_n (V_{GS} - V_{TH})^2, & V_{GS} > V_{TH} > V_{DS} \end{cases}$$

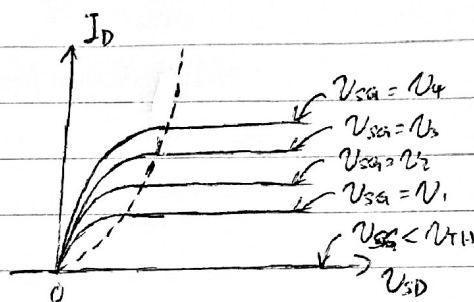
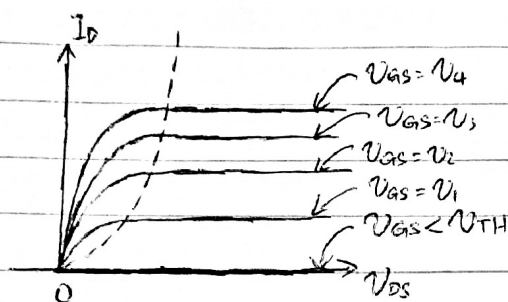
PMOS



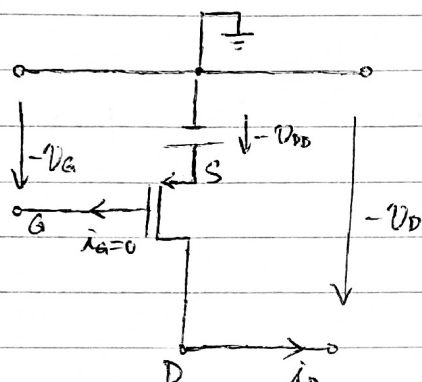
$$\lambda_G = 0$$

$$I_D = \begin{cases} 0, & V_{SG} < V_{TH} \\ 2\beta_p [(V_{SG} - V_{TH}) - 0.5 V_{SD}^2], & \min(V_{SG}, V_{SD}) > V_{TH} \\ \beta_p (V_{SG} - V_{TH})^2, & V_{SG} > V_{TH} > V_{SD} \end{cases}$$

伏安特性



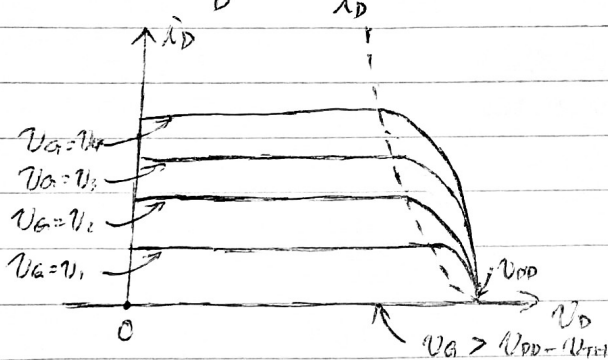
(6)



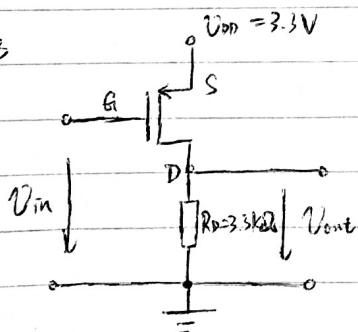
$$V_{SG} = -V_G - (-V_{DD}) = -V_G + V_{DD}$$

$$V_{SD} = -V_D + V_{DD}$$

$$I_D = \begin{cases} 0, & V_G > V_{DD} - V_{TH} \\ 2\beta_p [(V_{DD} - V_G - V_{TH})(V_{DD} - V_D) - 0.5(V_{DD} - V_D)^2], & \min(V_{DD} - V_G, V_{DD} - V_D) > V_{TH} \\ \beta_p (V_{DD} - V_G - V_{TH})^2, & V_{DD} - V_G > V_{TH} > V_{DD} - V_D \end{cases}$$



作第3



$$V_{out} = R_D I_D = \begin{cases} 0, & V_{DD} - V_{in} < V_{THp} \\ 2\beta_p [(V_{DD} - V_{in} - V_{THp})(V_{DD} - V_{out}) - 0.5(V_{DD} - V_{out})^2], & V_{DD} - V_{in} > V_{THp} \\ \beta_p (V_{DD} - V_{in} - V_{THp})^2, & V_{DD} - V_{in} > V_{THp} > V_{out} - V_{in} \end{cases}$$

當  $V_{in} > V_{DD} - V_{THp} = 2.6V$  時,  $I_D = 0A$ ,  $V_{out} = 0V$

當  $V_{in} < V_{DD} - V_{THp} = 2.6V$  且  $\beta_p (V_{DD} - V_{in} - V_{THp})^2 - V_{in} < V_{THp}$  時

$$\text{即 } 3.3 \cdot (V_{in} - 2.6)^2 - (V_{in} - 2.6) - 0.7 - 2.6 < 0 \text{ 時}$$

$$3.3(V_{in} - 2.6)^2 - 10(V_{in} - 2.6) - 3.3 < 0$$

$$\text{當 } 2.6 \pm \frac{10 - \sqrt{445}}{6.6} < V_{in} < 2.6 \text{ 時, 即 } 1.74 < V_{in} < 2.6 \text{ 時, } V_{out} = 3.3 \times (V_{in} - 2.6)^2 V$$

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當  $V_{in} < 1.74V$  時

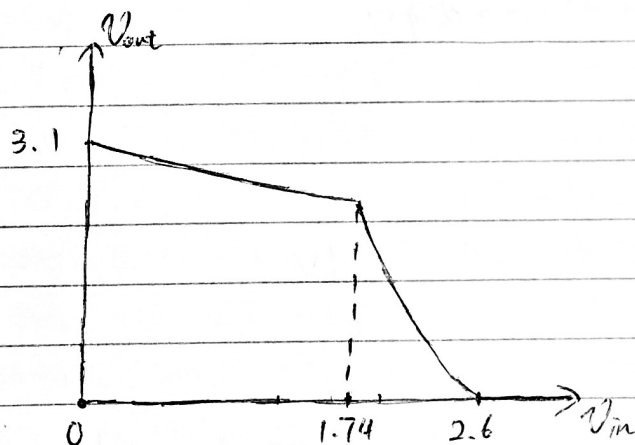
$$V_{out} = R_D \bar{I}_D = R_D \beta_p [(V_{DD} - V_{THP} - V_{in})(V_{DD} - V_{out}) - 0.5(V_{DD} - V_{out})^2] \times 2$$

$$= 3.3 \times [-0.5(V_{out} - 3.3)^2 + (V_{in} - 2.6)(V_{out} - 3.3)]$$

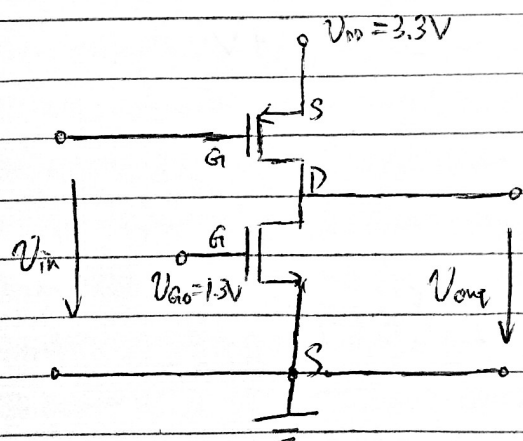
$$\Rightarrow 6.6(V_{in} - 2.6)(V_{out} - 3.3)$$

$$[6.6(V_{in} - 2.6) - 1] V_{out} = 6.6(V_{in} - 2.6) \times 3.3$$

$$V_{out} = \frac{3.3 \times 6.6(2.6 - V_{in})}{6.6(2.6 - V_{in}) + 1} = 3.3 - \frac{3.3}{6.6 \times (2.6 - V_{in}) + 1}$$



圖(a)的電壓轉移曲線

(1) 當PMOS I 作在截止區時,  $V_{DD} - V_{in} < V_{THP}$ 

$$\Leftrightarrow V_{in} > V_{DD} - V_{THP} = 2.6V$$

$$I_D = 0 \Rightarrow 2\beta_n [V_{DD} V_{out} - 0.5 V_{out}^2] = 0 \quad \text{且} \quad V_{out} < V_{DD}$$

$$\Rightarrow V_{out} = 0V$$

(2) 當PMOS I 作在恆流區時,  $V_{in} < 2.6V$ 

$$V_{out} - V_{in} < V_{THP} = 0.7V$$

$$I_D = \beta_p (V_{DD} - V_{in} - V_{THP})^2 = (2.6 - V_{in})^2 \text{ (mA)}$$

(2.1)

当 NMOS 管工作在恒流区时,  $V_{out} > V_{od} = 0.5V$

$$I_D = \beta_n V_{od}^2 = 0.625 \text{ mA}$$

$$\text{所以 } (2.6 - V_{in})^2 = 0.625$$

$$\Leftrightarrow 2.6 - V_{in} = \frac{\sqrt{10}}{4} \Leftrightarrow V_{in} = 2.6 - \frac{\sqrt{10}}{4} = 1.81V$$

$$\text{此时, } 0.5V < V_{out} < 0.7V + V_{in} = 2.51V$$

(2.2)

当 NMOS 管工作在欧姆区时,  $V_{out} < V_{od} = 0.5V$

$$I_D = 2\beta_n [V_{od} V_{out} - 0.5 V_{out}^2] \approx 2\beta_n V_{od} V_{out} = (2.6 - V_{in})^2$$

$$V_{out} = \frac{(2.6 - V_{in})^2}{2.5 \times 10^{-3}} = 400(2.6 - V_{in})^2 < 0.5V$$

$$2.6 - V_{in} < \frac{\sqrt{0.5}}{20} \Leftrightarrow V_{in} > 2.6 - \frac{\sqrt{0.5}}{20} = 2.56V$$

$$2.49V < V_{in} < 2.6V$$

(3) 当 PMOS 工作在欧姆区时,  $V_{in} < 2.6V$ ,  $V_{out} - V_{in} > 0.7V$

$$I_D \approx 2\beta_p (2.6 - V_{in})(3.3 - V_{out})$$

(3.1) 当 NMOS 工作在恒流区时,  $V_{out} > 0.5V$

$$I_D = 0.625 \text{ mA}$$

$$3.3 - \frac{0.3125}{2.6 - V_{in}} - V_{in} > 0.7$$

$$\Rightarrow 3.3 - V_{out} = \frac{I_D}{2\beta_p} \cdot \frac{1}{2.6 - V_{in}} = \frac{0.3125}{2.6 - V_{in}}$$

$$V_{in} < 2.04V$$

$$\Leftrightarrow V_{out} = 3.3 - \frac{0.3125}{2.6 - V_{in}} > 0.5$$

$$\frac{1}{2.8} < \frac{2.6 - V_{in}}{0.3125} \Leftrightarrow V_{in} < 2.6 - \frac{0.3125}{2.8} \approx 2.49V$$

(3.2) 当 NMOS 工作在欧姆区时,  $V_{out} < 0.5V$

$$I_D = 2\beta_p (2.6 - V_{in})(3.3 - V_{out}) = 2\beta_n V_{od} V_{out}$$

$$3.3 \times (2.6 - V_{in}) = [(2.6 - V_{in}) + 2.5 \times 0.5] V_{out}$$

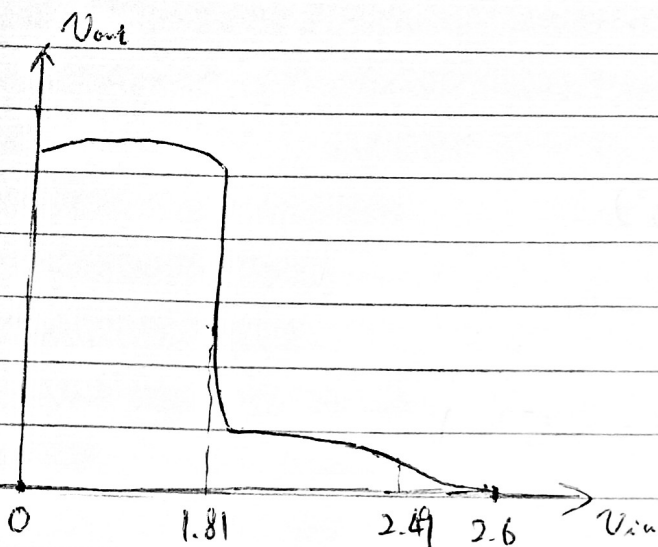
$$\Leftrightarrow V_{out} = \frac{3.3 \times (2.6 - V_{in})}{(2.6 - V_{in}) + 1.25} = 3.3 - \frac{4.125}{3.85 - V_{in}} < 0.5$$

$$\Leftrightarrow V_{in} < 2.38V$$



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#### 作業 4

(1) 對於無負反饋放大器，

$$V_{SG} = V_{DD} \cdot \frac{R_1}{R_1 + R_2} = 3.3 \times \frac{14.3k}{33k} = 1.43V > 0.8V = V_{TH}$$

$$V_{SD} = V_{DD} - V_L = 3.3V - I_{out} R_L$$

$$I_{out} = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{SG} - V_{TH})^2 = 0.99mA \approx 1mA$$

對於負反饋放大器

$$V_{SG} = V_{DD} \cdot \frac{R_1}{R_1 + R_2} = 1.93V > 0.8V = V_{TH}$$

$$I_{out} = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{DD} - I_{out} R_S - V_{TH})^2$$
$$= 25 \times 10^{-6} \times (1.13 - 500 I_{out})^2$$

$$400 I_{out} = 500^2 I_{out} - 1130 I_{out} + 1.2769$$

$$\Leftrightarrow 500^2 I_{out} - 1530 I_{out} + 1.2769 = 0$$

$$I_{out} = \frac{1530 \pm \sqrt{1064000}}{2 \times 500^2} \approx \frac{1530 \pm 1031.5}{2 \times 500^2}$$

$$I_{out} = 0.987mA \approx 1mA$$

(2) 對於負反饋電路,

$$I_D = \frac{1}{2} \mu_p C_{ox} \cdot \frac{W}{L} \times (1.13 - 500 I_D)^2$$

$$= \frac{1}{2} \times 95 \mu \times 50 \times (1.13 - 500 I_D)^2$$

$$\Leftrightarrow I_D = 2.375 \text{ mA} \times (500^2 I_D^2 - 1150 I_D + 1.13^2)$$

$$\Leftrightarrow 593.75 I_D^2 - 3.6875 I_D + 3.03 \times 10^{-3} = 0$$

$$I_D = \frac{3.68 \pm 2.52}{2 \times 593.75} \approx 9.77 \times 10^{-4} \text{ A} = 0.977 \text{ mA}$$

對於三負反饋電路

$$I_D = \frac{1}{2} \mu_p C_{ox} \cdot \frac{W}{L} \cdot (V_{GS} - V_{TH})^2$$

$$= 1 \text{ mA} \times 95\% = 0.95 \text{ mA}$$

所以負反饋更接近理想設計